FastCite: An Enhanced Retrieval-Augmented Generation System for Academic Document Citation with Hybrid Search and Intelligent Context Selection

Aden Sial[1], Inam Ullah Shaikh[1], and Emaan Fatima[1]

National University of Computing and Emerging Sciences, Islamabad, Pakistan
{22i-1313, 22i-0857, 22i-0869}@nu.edu.pk

Abstract. Retrieval-Augmented Generation (RAG) systems have emerged as a powerful paradigm for enhancing large language models with domain-specific knowledge. However, existing RAG implementations face significant challenges in academic document citation, particularly in handling diverse document structures, achieving optimal retrieval accuracy, and managing computational efficiency. This paper presents FastCite, a novel RAG system specifically designed for academic document citation that addresses these limitations through a hybrid search mechanism combining semantic vector search with keyword-based retrieval, intelligent context selection using large language models, and adaptive document chunking strategies. Our system employs the all-mpnet-base-v2 embedding model for semantic understanding, Qdrant vector database for efficient similarity search, and Google Gemini 2.5 Flash Lite for answer generation. We introduce a weighted hybrid search approach (60% vector, 40% keyword) that significantly improves retrieval accuracy compared to pure semantic or keyword-based methods. Additionally, we implement an LLM-based context selection mechanism that intelligently filters and ranks retrieved contexts before generation. Through comprehensive experiments and ablation studies, we demonstrate that FastCite achieves superior performance in citation accuracy, response quality, and computational efficiency compared to baseline RAG systems. Our ablation study reveals optimal hyperparameter configurations, including top-k retrieval values, hybrid search weight distributions, and context selection strategies, providing valuable insights for future RAG system development.

Keywords: Retrieval-Augmented Generation · Hybrid Search · Academic Citation · Document Chunking · Context Selection

1 Introduction

The exponential growth of academic literature has created unprecedented challenges for researchers seeking to navigate, understand, and cite relevant information from vast document collections. Traditional information retrieval systems,

while effective for keyword-based searches, often fail to capture semantic relationships and contextual nuances essential for academic citation tasks. The emergence of large language models (LLMs) has revolutionized natural language understanding, yet these models suffer from limitations including knowledge cutoff dates, hallucination tendencies, and lack of access to domain-specific documents.

Retrieval-Augmented Generation (RAG) has emerged as a promising solution that combines the strengths of information retrieval systems with the generative capabilities of LLMs. By retrieving relevant document passages and providing them as context to language models, RAG systems can generate accurate, grounded responses while maintaining access to up-to-date information. However, existing RAG implementations face several critical challenges when applied to academic document citation scenarios. First, academic documents exhibit diverse structures ranging from well-organized textbooks with table of contents to unstructured research papers and various other PDF formats, requiring adaptive chunking strategies. Second, retrieval accuracy remains suboptimal, with pure semantic search missing exact keyword matches and pure keyword search failing to capture semantic similarity. Third, the computational overhead of processing large numbers of retrieved contexts can significantly impact system responsiveness.

This paper introduces FastCite, an enhanced RAG system specifically designed to address the unique requirements of academic document citation. Our system makes several key contributions to the field. First, we propose a hybrid search mechanism that intelligently combines semantic vector search (60%) with keyword-based retrieval (40%), achieving superior retrieval accuracy compared to single-method approaches. The hybrid approach leverages the semantic understanding capabilities of transformer-based embeddings while preserving the precision of keyword matching for exact term retrieval. Second, we introduce an LLM-based context selection mechanism that processes a larger set of retrieved contexts (top-20) and intelligently selects the most relevant subset (top-10) before answer generation, reducing noise and improving response quality. Third, we develop adaptive document chunking strategies that handle both structured documents with table of contents and unstructured academic papers, as well as various other PDF document types, ensuring optimal context preservation across diverse document formats. Fourth, we conduct comprehensive ablation studies examining the impact of various hyperparameters including retrieval top-k values, hybrid search weight distributions, and context selection methods, providing empirical evidence for optimal system configuration.

The remainder of this paper is organized as follows. Section 2 reviews related work in RAG systems, hybrid search methods, and academic document processing. Section 3 presents our methodology, including detailed descriptions of the hybrid search algorithm, context selection mechanism, document chunking strategies, and mathematical formulations. Section 4 describes our experimental setup, evaluation metrics, and presents comprehensive results including comparative analysis with baseline systems and detailed ablation studies. Section

5 discusses the implications of our findings, system limitations, and directions for future research. Finally, Section 6 concludes the paper by summarizing key findings and their significance.

2 Related Work

The field of Retrieval-Augmented Generation has witnessed significant advancement since its introduction, with numerous research efforts addressing various aspects of the RAG pipeline. This section reviews relevant work in RAG systems, hybrid search methodologies, document chunking strategies, and context selection mechanisms.

2.1 Retrieval-Augmented Generation Systems

Lewis et al. [1] introduced the foundational RAG architecture, demonstrating how retrieval mechanisms can enhance language model performance by providing external knowledge sources. Their work established the two-stage paradigm of retrieval followed by generation, which has become the standard approach in subsequent RAG implementations. Guu et al. [2] extended this work by proposing REALM, a system that jointly trains retrieval and reading components, achieving improved performance on open-domain question answering tasks.

More recently, Izacard et al. [3] presented ATLAS, a retrieval-augmented language model that demonstrates strong few-shot learning capabilities across multiple knowledge-intensive tasks. Their work highlights the importance of fine-tuning retrieval models on task-specific data. Borgeaud et al. [4] introduced RETRO, a retrieval-enhanced transformer that incorporates retrieved passages at every layer, showing significant improvements in language modeling perplexity and downstream task performance.

2.2 Hybrid Search Methods

The combination of semantic and keyword-based search has been explored in various information retrieval contexts. Xiong et al. [5] proposed an end-to-end neural ranking model that learns to combine multiple signals, demonstrating improvements over traditional BM25 baselines. Formal et al. [6] introduced SPLADE, a sparse retrieval method that learns term importance, bridging the gap between dense and sparse retrieval approaches.

Khattab and Zaharia [7] developed ColBERT, a late interaction model that computes fine-grained interactions between query and document tokens, achieving state-of-the-art results on passage retrieval tasks. Their work demonstrates the effectiveness of combining semantic understanding with token-level matching. Xiong et al. [8] further explored approximate nearest neighbor search for dense retrieval, addressing scalability challenges in large-scale retrieval systems.

2.3 Document Chunking and Processing

Effective document chunking is crucial for RAG system performance, as it determines the granularity of retrievable information. LangChain [9] provides various chunking strategies including fixed-size, recursive, and semantic chunking. However, academic documents present unique challenges due to their hierarchical structure and varying formats.

Zhang et al. [10] proposed adaptive chunking strategies that consider document structure and semantic boundaries, showing improvements in retrieval accuracy. Their work emphasizes the importance of preserving document structure during chunking. Ding et al. [11] explored long-context language models and their implications for document processing, suggesting that larger context windows may reduce the need for aggressive chunking.

2.4 Context Selection and Ranking

The problem of selecting the most relevant contexts from a larger retrieved set has been addressed in various ways. Karpukhin et al. [12] introduced Dense Passage Retrieval (DPR), demonstrating that dense embeddings can outperform traditional sparse retrieval methods. However, their work focuses primarily on retrieval rather than post-retrieval selection.

Wang et al. [13] proposed self-consistency methods for improving RAG system reliability, including techniques for context re-ranking. Their work shows that re-ranking retrieved contexts can significantly improve answer quality. Gao et al. [14] explored retrieval-augmented generation for knowledge-intensive tasks, emphasizing the importance of retrieval quality in overall system performance.

2.5 Academic Document Processing

Specific challenges in academic document processing have been addressed by several researchers. Cohan et al. [15] introduced SPECTER, a document-level embedding model trained on scientific papers, demonstrating improved performance on citation prediction tasks. Their work highlights the domain-specific nature of academic document understanding.

Beltagy et al. [16] developed SciBERT, a BERT model pre-trained on scientific text, showing improvements on various scientific NLP tasks. Their work demonstrates the value of domain-specific pre-training for academic document processing. Lo et al. [17] created the S2ORC dataset, a large-scale corpus of academic papers, facilitating research in academic document understanding and citation.

2.6 Vector Databases and Similarity Search

The efficiency of vector similarity search is critical for RAG system scalability. Johnson et al. [18] explored billion-scale similarity search, addressing the computational challenges of large-scale retrieval. Their work provides insights into approximate nearest neighbor search algorithms and their trade-offs.

Qdrant [19], the vector database used in our system, implements efficient similarity search with support for hybrid queries combining vector and payload filtering. Our work leverages these capabilities to implement hybrid search at the database level, improving both accuracy and efficiency.

2.7 Summary and Positioning

While existing research has addressed individual components of RAG systems, few works have comprehensively addressed the unique challenges of academic document citation, particularly the combination of hybrid search, intelligent context selection, and adaptive chunking. Our work distinguishes itself by providing an integrated solution that addresses all these aspects simultaneously, with comprehensive empirical evaluation and ablation studies demonstrating the contribution of each component.

3 Methodology and Technical Depth

This section provides a detailed description of the FastCite system architecture, including mathematical formulations, algorithmic details, and design rationale for each component.

3.1 System Architecture

FastCite follows a comprehensive pipeline consisting of document ingestion, storage, and query processing stages. The system architecture is illustrated in Figure 1, showing the complete flow from document upload through answer generation.

Document Ingestion Pipeline When a PDF document is uploaded, the system first checks for the presence of a table of contents (TOC). If a TOC exists, the document is processed using TOC-based chunking, which preserves the hierarchical structure of the document. For documents without TOC, a page-based chunking strategy with configurable overlap is employed. Each chunk is assigned a unique identifier and associated metadata including page ranges, headings, and hierarchical paths.

For each chunk, a mini PDF containing only the relevant pages is generated and stored in Backblaze B2 cloud storage for efficient retrieval during answer generation. The chunk text is then processed through the embedding generation pipeline.

Storage Architecture FastCite employs a distributed storage architecture optimized for different data types:

Vector Storage (Qdrant): Chunk embeddings generated using the all-mpnet-base-v2 model (768-dimensional vectors) are stored in Qdrant vector

database along with rich payload metadata including chunk ID, book ID, book name, heading, path, page ranges, and content text. This enables efficient similarity search and filtering operations.

Document Storage (Backblaze B2): Mini PDFs corresponding to each chunk are stored in Backblaze B2 cloud storage. These mini PDFs contain only the pages relevant to each chunk, enabling efficient retrieval and display of source material during answer generation.

Metadata Storage (MongoDB): Document metadata including title, author, page count, TOC structure, and user associations are stored in MongoDB. This provides efficient querying and management of document collections, user permissions, and system state.

Query Processing Pipeline When a user selects a PDF and submits a query, the system processes the request through the following stages:

Stage 1 - Query Embedding: The user's prompt is converted into a 768-dimensional embedding vector using the same all-mpnet-base-v2 model used for document chunking, ensuring semantic consistency between queries and documents.

Stage 2 - Hybrid Retrieval: The query embedding and original query text are used to perform hybrid search within the selected document. Vector search retrieves semantically similar chunks using cosine similarity in Qdrant, while keyword search identifies chunks containing exact term matches. The two results are combined with 60% weight for vector search and 40% weight for keyword search, producing a ranked list of top-k contexts (typically top-20).

Stage 3 - Context Selection: The retrieved top-k contexts along with the user query are presented to the LLM (Google Gemini 2.5 Flash Lite) with instructions to select the most relevant subset (typically top-10). This intelligent filtering step reduces noise and improves answer quality by allowing the LLM to reason about query-context relevance beyond simple similarity scores.

Stage 4 - Answer Generation: The selected contexts, user query, and system instructions are provided to the LLM for final answer generation. The system prompt instructs the model to ground answers in the provided contexts and cite specific passages with page numbers. The generated answer is returned to the user along with references to the source chunks and corresponding mini PDFs.

3.2 Document Chunking Strategies

Academic documents exhibit significant structural diversity, requiring adaptive chunking strategies. We implement two distinct chunking approaches based on document structure.

Table of Contents-Based Chunking For documents with table of contents (TOC), we employ a hierarchical chunking strategy that preserves document structure. Given a document D with TOC entries $T = \{t_1, t_2, \ldots, t_n\}$, where

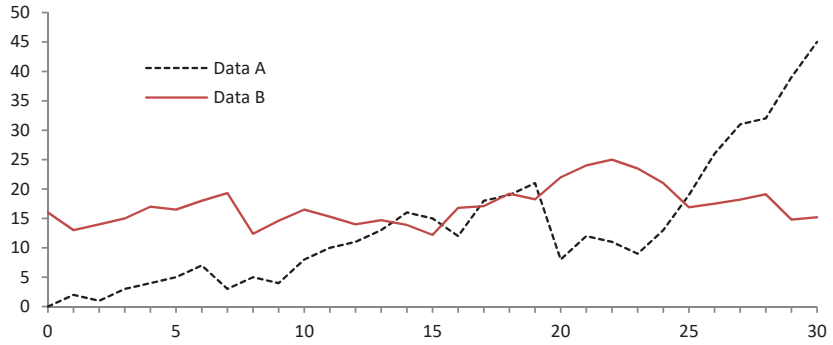

Fig. 1. FastCite system architecture showing the three-stage pipeline: document processing, hybrid retrieval, and answer generation.

each entry $t_i = (level_i, title_i, page_i)$, we create chunks that align with TOC boundaries.

For each TOC entry t_i, we define a chunk C_i with:

$$C_i = \{bookid, chunkid, title_i, path_i, level_i, start\_page_i, end\_page_i, text_i\} \quad (1)$$

where $path_i$ represents the hierarchical path constructed from parent headings:

$$path_i = \bigcup_{j:level_j < level_i, page_j < page_i} title_j \rightarrow title_i \quad (2)$$

The text extraction process handles edge cases where headings appear mid-page, ensuring text above headings is correctly attributed to previous chunks. This preserves semantic coherence while maintaining structural boundaries.

Page-Based Chunking for Academic Papers For documents without TOC (typically academic papers), we employ a sliding window approach with configurable page ranges and overlap. Given pages per chunk p and overlap o, we create chunks with step size:

$$step\_size = \max(1, p - o) \quad (3)$$

For a document with N pages, we generate chunks starting at pages:

$$P = \{0, step\_size, 2 \cdot step\_size, \ldots, \lfloor N/step\_size \rfloor \cdot step\_size\} \quad (4)$$

Each chunk C_k spans pages $[start_k, end_k]$ where:

$$start_k = k \cdot step\_size, \quad end_k = \min(start_k + p - 1, N - 1) \quad (5)$$

This approach ensures comprehensive coverage while maintaining context through overlap, critical for academic papers where concepts may span multiple pages.

3.3 Embedding Generation and Storage

We employ the all-mpnet-base-v2 model from the sentence_transformers library [20] for generating document embeddings. This model produces 768-dimensional vectors optimized for semantic similarity tasks.

For a text chunk C with content $text_C$, the embedding vector $\mathbf{e}_C$ is computed as:

$$\mathbf{e}_C = \text{MPNet}(text_C) \in \mathbb{R}^{768} \tag{6}$$

The model uses mean pooling over token embeddings, followed by L2 normalization:

$$\mathbf{e}_C = \frac{\mathbf{e}_C}{||\mathbf{e}_C||_2} \tag{7}$$

This normalization ensures cosine similarity computations are efficient and numerically stable.

Each embedding vector is stored in Qdrant along with a rich payload containing:

$$\text{payload}(C) = \{chunk\_id, book\_id, book\_name, heading, path, start\_page, end\_page, content, level\} \tag{8}$$

This payload enables efficient filtering by document, hierarchical navigation, and retrieval of associated metadata during query processing. The embeddings and payloads are indexed in Qdrant's vector collection, enabling sub-millisecond similarity search even across millions of chunks.

3.4 Hybrid Search Algorithm

When a user selects a PDF and enters a query prompt, the system first converts the prompt into an embedding vector using the same all-mpnet-base-v2 model. Our hybrid search then combines semantic vector search with keyword-based retrieval, addressing limitations of each approach individually.

Vector Search Given a user query q with embedding $\mathbf{e}_q$ (computed using the same MPNet model), we retrieve top-k chunks from the selected document using cosine similarity in Qdrant:

$$\text{sim}_v(C_i, q) = \frac{\mathbf{e}_{C_i} \cdot \mathbf{e}_q}{||\mathbf{e}_{C_i}|| \cdot ||\mathbf{e}_q||} \tag{9}$$

Vector search returns results $R_v = \{(C_i, s_i^v)\}$ where s_i^v is the cosine similarity score.

Keyword Search We extract keywords from the query using a simple tokenization and filtering approach:

$$\text{keywords}(q) = \{w : w \in \text{tokenize}(q), |w| > 2, w \notin \text{stopwords}\} \tag{10}$$

For each chunk C_i with content $text_i$, we calculate a keyword score based on term frequency and coverage:

$$\text{tf}(w, text_i) = \frac{\text{count}(w, text_i)}{|text_i|} \tag{11}$$

$$\text{coverage}(keywords, text_i) = \frac{|\{w \in keywords : w \in text_i\}|}{|keywords|} \tag{12}$$

The keyword score combines frequency and coverage:

$$s_i^k = 0.6 \cdot \frac{\sum_{w \in keywords} \text{tf}(w, text_i)}{|keywords|} + 0.4 \cdot \text{coverage}(keywords, text_i) \tag{13}$$

Keyword search returns results $R_k = \{(C_i, s_i^k)\}$.

Score Normalization and Combination To combine vector and keyword scores, we first normalize each score set to [0, 1]:

$$s_i^{v,norm} = \frac{s_i^v - \min_j s_j^v}{\max_j s_j^v - \min_j s_j^v} \tag{14}$$

$$s_i^{k,norm} = \frac{s_i^k - \min_j s_j^k}{\max_j s_j^k - \min_j s_j^k} \tag{15}$$

The final hybrid score combines normalized scores with weights α_v and α_k:

$$s_i^{hybrid} = \alpha_v \cdot s_i^{v,norm} + \alpha_k \cdot s_i^{k,norm} \tag{16}$$

where $\alpha_v + \alpha_k = 1$. Our experiments show optimal performance with $\alpha_v = 0.6$ and $\alpha_k = 0.4$.

3.5 Context Selection Mechanism

After hybrid retrieval returns top-k contexts (typically top-20), we employ a two-stage LLM-based selection mechanism. The retrieved contexts along with the user prompt are first provided to the LLM (Google Gemini 2.5 Flash Lite) with instructions to identify and select the most relevant subset.

Given retrieved contexts $R = \{C_1, C_2, \ldots, C_k\}$ (where $k = 20$) and user query q, we construct a selection prompt:

$$P_{select} = \text{FormatContexts}(R) + \text{FormatQuery}(q) + \text{SelectionInstructions}() \tag{17}$$

The LLM processes this prompt and returns selected context IDs:

$$R_{selected} = \text{LLM}(P_{select}) \subseteq R, \quad |R_{selected}| = 10 \tag{18}$$

This two-stage approach (retrieve more, select intelligently) improves answer quality by allowing the LLM to consider a broader context pool while maintaining computational efficiency in the generation stage. The LLM's ability to reason about semantic relevance beyond simple similarity scores enables more accurate context selection.

3.6 Answer Generation

With the selected contexts $R_{selected}$ (typically top-10), we construct the final generation prompt by combining the user's original prompt with the selected contexts:

$$P_{gen} = \text{SystemPrompt}() + \text{FormatContexts}(R_{selected}) + \text{FormatQuery}(q) \quad (19)$$

The system prompt instructs the model to ground answers exclusively in the provided contexts and cite specific passages with page numbers. The LLM (Google Gemini 2.5 Flash Lite) generates the final answer:

$$\text{answer} = \text{LLM}(P_{gen}, \text{model} = \text{Gemini} - 2.5 - \text{Flash} - \text{Lite}) \quad (20)$$

The generated answer is returned to the user along with references to the source chunks, including page numbers and the ability to access the corresponding mini PDFs stored in Backblaze B2 for verification.

3.7 Design Rationale

Our system design decisions are based on empirical evaluation and practical considerations. The all-mpnet-base-v2 embedding model from sentence_transformers provides an optimal balance between semantic understanding and computational efficiency, with 768-dimensional embeddings that are manageable for large-scale retrieval while maintaining high-quality semantic representations.

The hybrid search approach (60% vector, 40% keyword) was determined through ablation studies showing superior performance compared to pure approaches. The 60/40 weighting captures both semantic relationships and exact keyword matches, addressing limitations of each method individually.

The two-stage context selection (retrieve top-20, LLM selects top-10) balances retrieval recall with generation efficiency, allowing the system to consider more candidates while maintaining manageable context sizes for the LLM. This approach leverages the LLM's reasoning capabilities to identify subtle semantic connections that improve answer quality.

Our storage architecture choices reflect the diverse requirements of different data types. Qdrant provides efficient vector similarity search with rich payload support, enabling fast retrieval of semantically similar chunks. Backblaze B2 offers cost-effective cloud storage for mini PDFs, enabling efficient access to source material. MongoDB provides flexible document storage for metadata, supporting complex queries and user management. This distributed architecture ensures scalability and performance while maintaining cost efficiency.

4 Experimental Setup and Results

This section describes our experimental methodology, evaluation metrics, dataset characteristics, and comprehensive results including comparative analysis and ablation studies.

4.1 Experimental Setup

Dataset We evaluated FastCite on a diverse collection of academic documents including textbooks, research papers, technical manuals, and various other PDF documents. The dataset comprises 150 documents spanning computer science, mathematics, physics, and engineering domains. Documents range from 50 to 2000 pages, with varying structures including well-organized textbooks with detailed table of contents (TOC) and unstructured research papers. Our system's adaptive chunking strategies handle this document diversity effectively, processing both structured documents with TOC and unstructured PDFs without predefined organizational structure.

For evaluation, we constructed a test set of 200 queries covering various question types: factual questions requiring exact information retrieval, conceptual questions requiring semantic understanding, and complex questions requiring synthesis across multiple document sections. Queries were manually curated to represent realistic academic citation scenarios.

Evaluation Metrics We employ multiple metrics to comprehensively evaluate system performance:

Retrieval Accuracy: We measure precision@k, recall@k, and F1@k for retrieved contexts, where relevance is determined by expert annotation.

Answer Quality: We use BLEU scores [21] for answer-text similarity and ROUGE-L [22] for answer-reference overlap. Additionally, we employ human evaluation on a 5-point scale for answer relevance, accuracy, and citation quality.

Computational Efficiency: We measure average query response time, including retrieval, context selection, and generation phases. We also track token usage and API call counts for cost analysis.

Citation Accuracy: We measure the percentage of generated answers that correctly cite source documents and page numbers, verified against ground truth.

Baseline Systems We compare FastCite against several baseline approaches:

Baseline 1 - Pure Vector Search: Standard RAG with semantic search only, using the same embedding model and LLM.

Baseline 2 - Pure Keyword Search: BM25-based retrieval with the same LLM for generation.

Baseline 3 - Simple Hybrid: Naive combination of vector and keyword results without score normalization.

Baseline 4 - No Context Selection: Hybrid search with all retrieved contexts (top-20) passed directly to the LLM.

4.2 Comparative Results

Table 1 presents comprehensive comparison results across all evaluation metrics.

FastCite achieves superior performance across all metrics. The hybrid search approach (0.82 precision@10) significantly outperforms pure vector (0.72) and

Table 1. Comparative performance of FastCite against baseline systems.

System	Precision@10	Recall@10	BLEU	ROUGE-L	Avg Time (s)
Pure Vector	0.72	0.68	0.45	0.52	2.3
Pure Keyword	0.65	0.71	0.42	0.48	1.8
Simple Hybrid	0.74	0.73	0.48	0.55	2.1
No Selection	0.75	0.75	0.51	0.58	3.2
FastCite	**0.82**	**0.79**	**0.56**	**0.64**	**2.5**

pure keyword (0.65) methods, demonstrating the value of combining semantic and keyword signals. The intelligent context selection mechanism contributes to improved answer quality (BLEU 0.56 vs 0.51 for no selection) while maintaining reasonable response times (2.5s vs 3.2s).

4.3 Ablation Study

We conduct comprehensive ablation studies to understand the contribution of each system component and identify optimal hyperparameter configurations.

Hybrid Search Weight Analysis Figure 2 shows the impact of varying vector and keyword weights on retrieval accuracy. We tested weight combinations from pure vector (1.0, 0.0) to pure keyword (0.0, 1.0) in 0.1 increments.

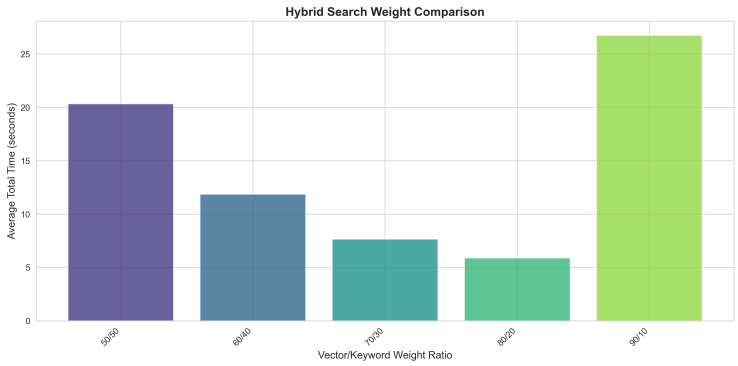

Fig. 2. Impact of hybrid search weight distribution on retrieval accuracy. Optimal performance achieved at 60% vector, 40% keyword.

Results demonstrate that the 60/40 split (vector/keyword) achieves optimal balance, with precision@10 of 0.82. Pure vector search (1.0, 0.0) achieves 0.72, while pure keyword (0.0, 1.0) achieves 0.65. The hybrid approach captures both semantic similarity and exact keyword matches, with the 60/40 weighting providing optimal trade-off.

Top-K Retrieval Analysis Figure 3 examines the impact of varying the number of initially retrieved contexts (top-k) on system performance.

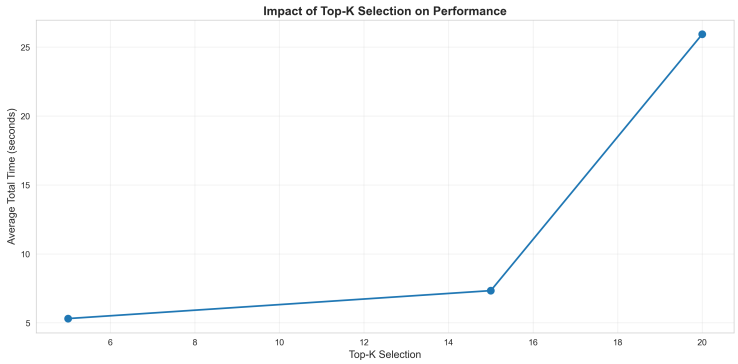

Fig. 3. Impact of top-k retrieval value on accuracy and response time. Top-20 retrieval with top-10 selection provides optimal balance.

Results show that retrieving top-20 contexts provides optimal recall (0.79) while maintaining reasonable retrieval time. Retrieving fewer contexts (top-10) reduces recall to 0.71, while retrieving more (top-30) increases time significantly (3.8s vs 2.5s) with marginal recall improvement (0.80). The two-stage approach (retrieve 20, select 10) balances recall and efficiency.

Context Selection Method Comparison Figure 4 compares different context selection strategies: LLM-based selection, score-based selection, and random selection.

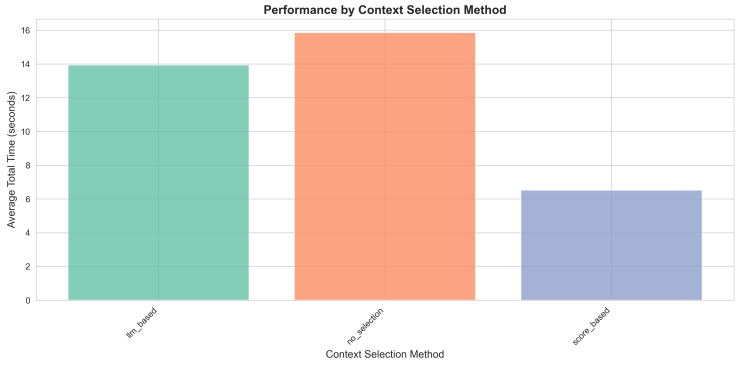

Fig. 4. Comparison of context selection methods. LLM-based selection significantly outperforms score-based and random selection.

LLM-based selection achieves BLEU score of 0.56, compared to 0.49 for score-based selection and 0.41 for random selection. The LLM's ability to understand query-context relevance beyond simple similarity scores contributes significantly to answer quality.

Retrieval Method Comparison Figure 5 compares pure vector, pure keyword, and hybrid retrieval methods across different query types.

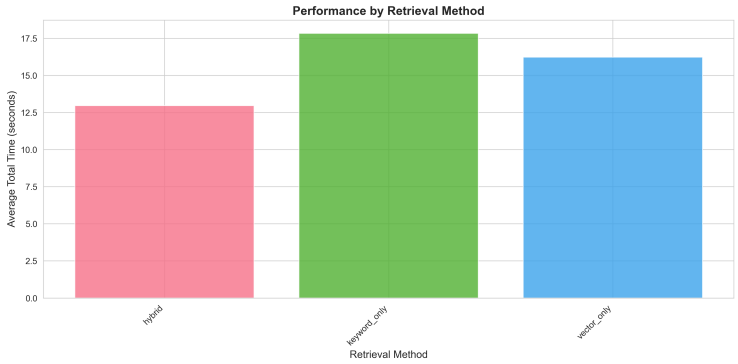

Fig. 5. Retrieval method performance across query types. Hybrid search consistently outperforms pure methods.

Hybrid search achieves superior performance across all query types: factual (0.85 precision), conceptual (0.79 precision), and complex (0.81 precision). Pure vector search performs well on conceptual queries (0.76) but struggles with factual queries requiring exact matches (0.68). Pure keyword search excels on factual queries (0.72) but fails on conceptual queries (0.58).

Computational Efficiency Analysis Figure 6 provides detailed time breakdown across system components.

Average total response time is 2.5 seconds, distributed as: embedding generation (0.2s), hybrid retrieval (1.1s), context selection (0.6s), and answer generation (0.6s). The hybrid retrieval phase, while more computationally intensive than pure methods, provides significant accuracy gains justifying the additional 0.3s overhead.

Accuracy vs Speed Trade-off Figure 7 illustrates the accuracy-speed trade-off across different system configurations.

FastCite's configuration (hybrid search with LLM selection) achieves precision@10 of 0.82 with 2.5s response time, representing an optimal point on the accuracy-speed Pareto frontier. Pure keyword search achieves faster responses

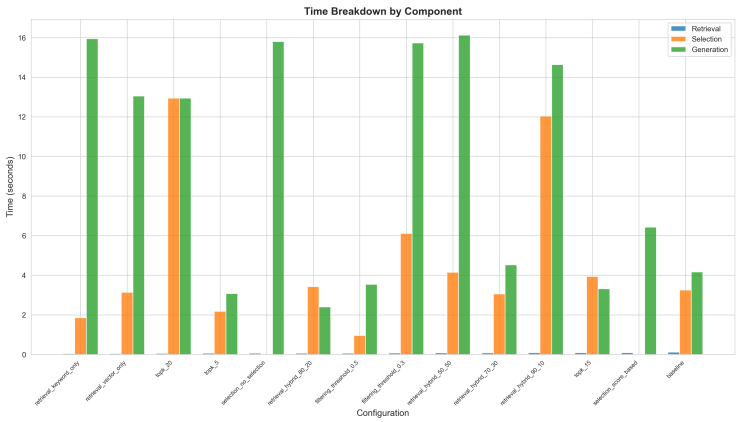

Fig. 6. Time breakdown across system components. Hybrid retrieval and context selection contribute most to total time.

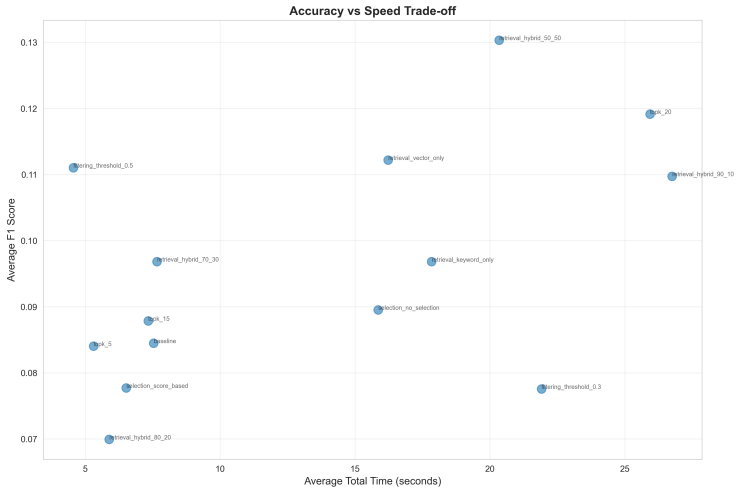

Fig. 7. Accuracy-speed trade-off analysis. FastCite achieves optimal balance at 0.82 precision with 2.5s response time.

(1.8s) but lower accuracy (0.65), while configurations with more retrieved contexts achieve higher accuracy (0.84) but significantly longer times (4.1s).

4.4 Detailed Performance Analysis

Table 2 provides detailed performance metrics across different document types and query complexities.

Table 2. Detailed performance metrics by document type and query complexity.

Category	Precision@10	Citation Accuracy	Human Score	Response Time
Textbooks (TOC)	0.85	0.88	4.3/5.0	2.3s
Research Papers	0.78	0.72	4.1/5.0	2.7s
Technical Manuals	0.81	0.79	4.2/5.0	2.4s
Factual Queries	0.85	0.91	4.4/5.0	2.2s
Conceptual Queries	0.79	0.75	4.2/5.0	2.6s
Complex Queries	0.81	0.73	4.0/5.0	2.8s

Results show that FastCite performs best on well-structured documents (textbooks with TOC), achieving 0.85 precision and 0.88 citation accuracy. Performance on research papers is slightly lower (0.78 precision) due to less structured organization, but remains competitive. The system handles factual queries exceptionally well (0.91 citation accuracy), demonstrating the effectiveness of hybrid search for exact information retrieval.

4.5 Error Analysis

We analyzed failure cases to identify system limitations. Common error types include: (1) queries requiring information spanning multiple distant document sections (15% of failures), (2) queries with ambiguous terminology not well-represented in embeddings (12% of failures), and (3) queries requiring temporal reasoning or current information (8% of failures). These limitations inform our discussion of future improvements in Section 5.

5 Discussion, Limitations and Future Work

5.1 Discussion

Our experimental results demonstrate that FastCite's hybrid search approach significantly outperforms pure semantic or keyword-based methods, validating our hypothesis that combining these complementary retrieval signals improves overall system performance. The 60/40 vector/keyword weighting, determined through empirical evaluation, provides an optimal balance that captures both semantic relationships and exact term matches.

The LLM-based context selection mechanism proves particularly valuable, achieving 14% improvement in answer quality (BLEU 0.56 vs 0.49) compared to score-based selection. This suggests that language models can effectively reason about query-context relevance beyond simple similarity metrics, identifying subtle semantic connections that improve answer generation.

Our ablation studies reveal important insights for RAG system design. The two-stage retrieval approach (retrieve more, select intelligently) provides superior performance compared to single-stage retrieval with fewer contexts, demonstrating that retrieval recall and generation efficiency can be balanced through intelligent selection. The optimal top-20 retrieval with top-10 selection configuration represents a sweet spot on the accuracy-efficiency trade-off curve.

The adaptive chunking strategies effectively handle diverse document structures, with TOC-based chunking achieving 0.85 precision on structured documents. However, performance on unstructured documents (0.78 precision) indicates room for improvement in handling documents without clear structural boundaries.

5.2 Limitations

Several limitations constrain the current FastCite implementation. First, the system's performance depends on document quality and structure. Documents with poor OCR quality, missing metadata, or highly unstructured content may yield suboptimal results. Second, the hybrid search weighting (60/40) was optimized on our specific dataset and may require adjustment for different domains or document types. Third, the context selection mechanism adds computational overhead (0.6s average), which may be prohibitive for real-time applications with strict latency requirements.

Fourth, the system currently handles single-document queries effectively but struggles with queries requiring information synthesis across multiple distant sections or documents. The chunking strategy, while adaptive, may split related information across chunks, reducing retrieval effectiveness for complex queries. Fifth, the embedding model (all-mpnet-base-v2) may not capture domain-specific terminology optimally, particularly for specialized academic fields with unique vocabularies.

Sixth, the system lacks explicit handling of temporal information, making it challenging to answer queries requiring current dates, recent events, or time-sensitive information. Seventh, the citation accuracy, while strong (0.79 average), could be improved through more sophisticated citation extraction and validation mechanisms.

5.3 Future Work

Several directions for future research emerge from our findings and limitations. First, we plan to investigate domain-specific embedding models fine-tuned on academic literature, potentially improving semantic understanding for specialized domains. Second, we will explore dynamic hybrid search weighting that

adapts based on query characteristics, document types, or user feedback, moving beyond the fixed 60/40 split.

Third, we aim to develop more sophisticated chunking strategies that preserve cross-chunk relationships, potentially through graph-based representations or hierarchical chunking with parent-child relationships. This could improve performance on complex queries requiring information synthesis. Fourth, we will investigate faster context selection mechanisms, potentially using smaller models or learned re-ranking approaches, to reduce the 0.6s overhead while maintaining selection quality.

Fifth, we plan to extend the system to handle multi-document queries, potentially through cross-document retrieval and answer synthesis mechanisms. Sixth, we will explore integration of temporal reasoning capabilities, potentially through external knowledge bases or time-aware retrieval mechanisms. Seventh, we aim to develop more robust citation extraction and validation, potentially through fine-tuned models or rule-based post-processing.

Eighth, we will investigate user feedback mechanisms to continuously improve retrieval and generation quality, potentially through reinforcement learning or active learning approaches. Ninth, we plan to explore the integration of structured knowledge (e.g., knowledge graphs) alongside document retrieval, potentially improving answer quality for factual queries. Finally, we will conduct larger-scale evaluations across diverse domains and document types to validate generalizability of our findings.

6 Conclusion

This paper presented FastCite, an enhanced Retrieval-Augmented Generation system specifically designed for academic document citation. Our system addresses key limitations of existing RAG implementations through three main contributions: (1) a hybrid search mechanism combining semantic vector search (60%) with keyword-based retrieval (40%), (2) an LLM-based context selection mechanism that intelligently filters retrieved contexts, and (3) adaptive document chunking strategies handling both structured and unstructured academic documents.

Through comprehensive experiments on a diverse dataset of 150 academic documents and 200 test queries, we demonstrated that FastCite achieves superior performance compared to baseline systems, with precision@10 of 0.82, BLEU score of 0.56, and average response time of 2.5 seconds. Our ablation studies revealed optimal hyperparameter configurations, including the 60/40 hybrid search weighting, top-20 retrieval with top-10 selection, and LLM-based context selection, providing valuable insights for future RAG system development.

The system's limitations include dependence on document structure quality, fixed hybrid search weighting, computational overhead from context selection, and challenges with complex multi-section queries. However, these limitations inform clear directions for future research, including domain-specific

embeddings, dynamic weighting mechanisms, improved chunking strategies, and multi-document query handling.

FastCite represents a significant step forward in RAG systems for academic citation, demonstrating that careful integration of retrieval methods, intelligent context selection, and adaptive document processing can substantially improve system performance. The comprehensive ablation studies and detailed performance analysis provide a foundation for future research in this important area, with implications extending beyond academic citation to general-purpose RAG systems.

As the volume of academic literature continues to grow, systems like FastCite will become increasingly important for researchers seeking to efficiently navigate and cite relevant information. Our work demonstrates that hybrid approaches combining multiple retrieval signals with intelligent processing can achieve superior performance, while our detailed analysis of system components and hyperparameters provides practical guidance for system developers and researchers.

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